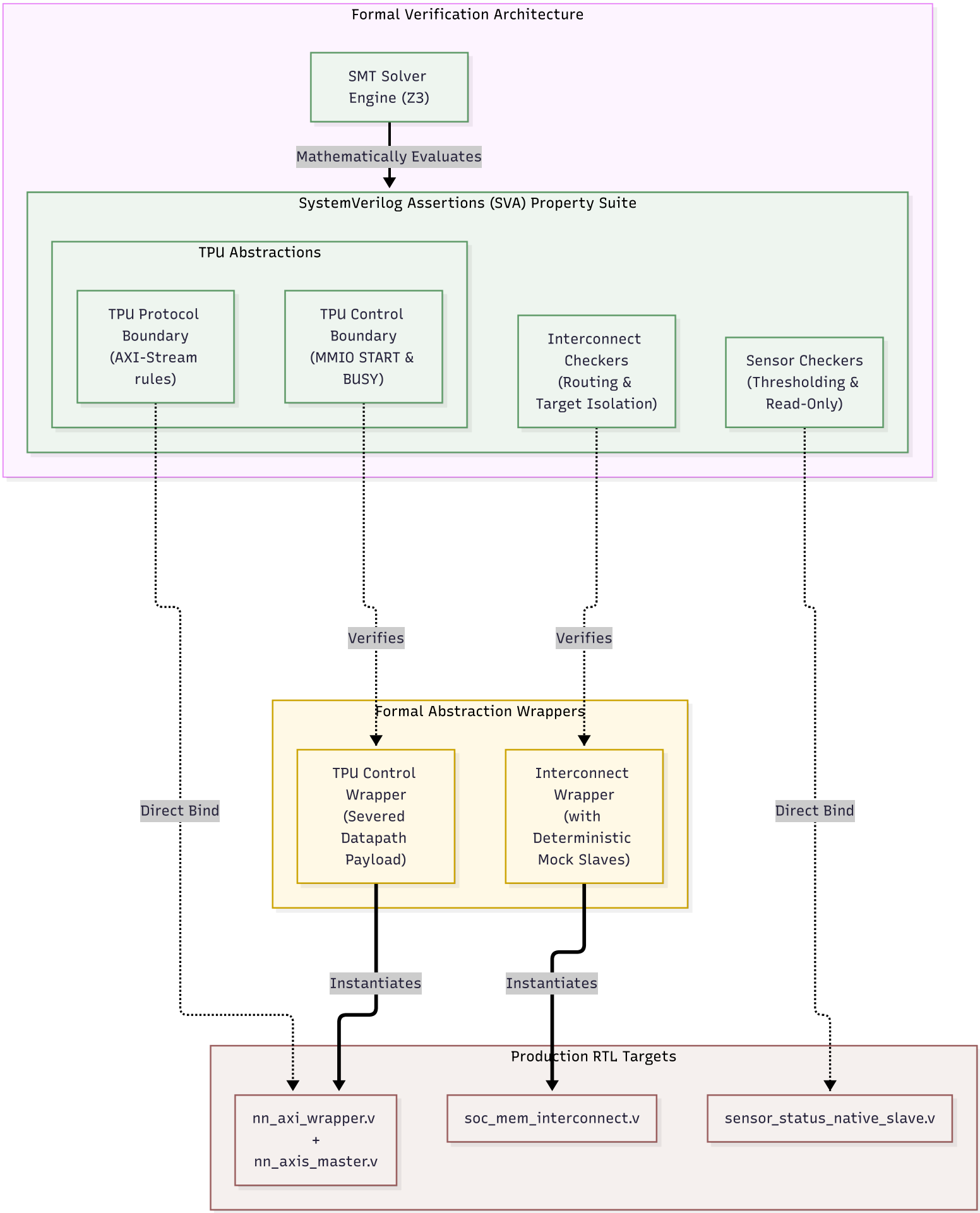

Formal Verification Architecture
SMT Solver Engine (Z3)
Mathematically Evaluates
SystemVerilog Assertions (SVA) Property Suite
TPU Abstractions
TPU Protocol Boundary (AXI-Stream rules)
TPU Control Boundary (MMIO START & BUSY)
Interconnect Checkers (Routing & Target Isolation)
Sensor Checkers (Thresholding & Read-Only)
Verifies
Verifies
Formal Abstraction Wrappers
TPU Control Wrapper (Severed Datapath Payload)
Interconnect Wrapper (with Deterministic Mock Slaves)
Direct Bind
Direct Bind
Instantiates
Instantiates
Production RTL Targets
nn_axi_wrapper.v + nn_axis_master.v
soc_mem_interconnect.v
sensor_status_native_slave.v